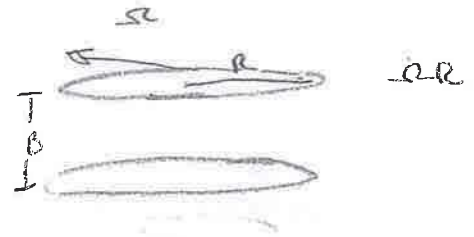


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TRANSPORT PHENOMENA I

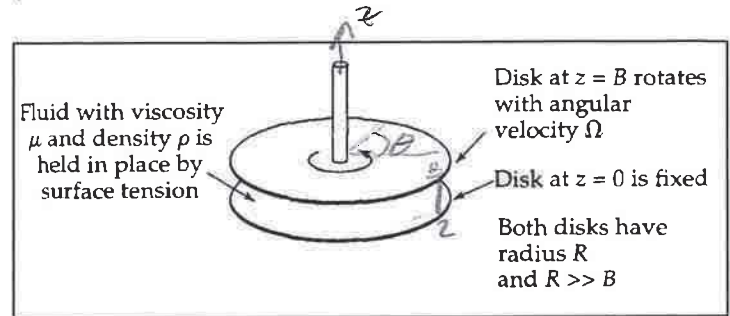
FALL 2025

EXAM 2



PARALLEL-DISK VISCOMETER

A fluid is placed in the gap between two horizontal circular disks of radius R . Disks are separated by a small distance B . The lower disk is held fixed. The upper disk is rotating at a constant angular velocity Ω .



Postulates:

- Newtonian fluid with constant density ρ and constant viscosity μ .
- $v_r = 0$ and $v_z = 0$. $v_\theta =$
- Pressure in the fluid is not a function of θ .
- Creeping flow assumption: The flow is very slow and the acceleration term $\frac{D(v)}{Dt}$ in the equation of motion is negligible.



1. Using the continuity equation, prove v_θ is not a function of θ at steady-state conditions.
2. Using the proper equation of motion at steady-state conditions, write down and simplify a partial differential equation for v_θ as a function of r and z .

3. Assuming v_θ may be written as r multiplied by a function of z ,

$$v_\theta(r, z) = r f(z) \rightarrow v_\theta(r, z) = r = f(z)$$

, convert the partial differential equation for v_θ to an ordinary differential equation for f .

4. Knowing $v_\theta(r, z = 0) = 0$ and $v_\theta(r, z = B) = r \Omega$, write down two boundary conditions for f , solve the ordinary differential equation derived in part 1.3. and evaluate the integration constants.

$$BC \begin{cases} f(z=B) = R \Omega \\ f(z=0) = 0 \end{cases}$$
5. One can estimate the viscosity of the fluid by measuring the torque T_z required to rotate the upper disk at a constant angular velocity. Using the velocity profile derived in part 1.4., find an equation for the viscosity in terms of T_z , Ω , B and R . Hint: torque exerted by the liquid on unit area of the upper disk is the product of the radius r (lever arm) and viscous stress in θ -direction.

$$\#4 \rightarrow f(z) BC \begin{cases} f(z=B) = R \Omega \\ f(z=0) = 0 \end{cases}$$

$$\text{Torque} = 2\pi r L$$

$$\text{Area} \times \tau_\theta \times r$$

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§B.1 NEWTON'S LAW OF VISCOSITY

$$[\tau = -\mu(\nabla \mathbf{v} + (\nabla \mathbf{v})^T) + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v})\delta]$$

Cylindrical coordinates (r, θ, z) :

$$\tau_{rr} = -\mu \left[2 \frac{\partial v_r}{\partial r} \right] + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v}) \quad (\text{B.1-8})^a$$

$$\tau_{\theta\theta} = -\mu \left[2 \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) \right] + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v}) \quad (\text{B.1-9})^a$$

$$\tau_{zz} = -\mu \left[2 \frac{\partial v_z}{\partial z} \right] + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v}) \quad (\text{B.1-10})^a$$

$$\tau_{r\theta} = \tau_{\theta r} = -\mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right] \quad (\text{B.1-11})$$

$$\tau_{\theta z} = \tau_{z\theta} = -\mu \left[\frac{1}{r} \frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \right] \quad (\text{B.1-12})$$

$$\tau_{zr} = \tau_{rz} = -\mu \left[\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right] \quad (\text{B.1-13})$$

in which

$$(\nabla \cdot \mathbf{v}) = \frac{1}{r} \frac{\partial}{\partial r} (rv_r) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \quad (\text{B.1-14})$$

^a When the fluid is assumed to have constant density, the term containing $(\nabla \cdot \mathbf{v})$ may be omitted. For monatomic gases at low density, the dilatational viscosity κ is zero.

§B.4 THE EQUATION OF CONTINUITY^a

$$[\partial \rho / \partial t + (\nabla \cdot \rho \mathbf{v}) = 0]$$

Cylindrical coordinates (r, θ, z) :

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial}{\partial z} (\rho v_z) = 0 \quad (\text{B.4-2})$$

§B.6 EQUATION OF MOTION FOR A NEWTONIAN FLUID WITH CONSTANT ρ AND μ

$$[\rho D\mathbf{v}/Dt = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g}]$$

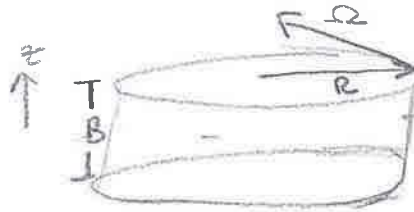
Cylindrical coordinates (r, θ, z) :

$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (rv_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r \quad (\text{B.6-4})$$

$$\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (rv_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta \quad (\text{B.6-5})$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \quad (\text{B.6-6})$$

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$\Omega = \text{const}$
 $N = \text{const}$
 $\rho = \text{const}$

IL $v_\theta(r)$

Egn of Continuity

$$\text{BC } \begin{cases} v_\theta(z=B, \theta) = R\Omega \\ v_\theta(z=0, \theta) = 0 \end{cases}$$

$[\partial \rho / \partial t + (\nabla \cdot \rho \mathbf{v}) = 0] \rightarrow$ were at steady state w/ const ρ so that term cancels out

$$\textcircled{1} \frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial}{\partial z} (\rho v_z) = 0$$

the fluid is traveling in the r direction & doesn't depend on θ changing so our v_θ wouldn't be a function of θ & z is changing the height which lets the flow through

$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + r g_\theta$$

$\leftarrow v_\theta(r, z)$

the $\frac{Dv_\theta}{Dt}$ is negligible

$$\rightarrow 0 = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{\partial^2 v_\theta}{\partial z^2} \right]$$

convert PDE $v_\theta(r, z) \rightarrow f(z)$

$$0 = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{\partial^2 v_\theta}{\partial z^2} \right]$$

$$-\mu \frac{\partial^2 v_\theta}{\partial z^2} = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) \right] \rightarrow -\frac{\partial^2 v_\theta}{\partial z^2} = \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right)$$

I need to get rid of $\frac{\partial v_\theta}{\partial r}$ to just leave z tried to separate variables but that makes no sense

$$-v_\theta + \frac{C_1}{r} + C_2 = 0$$

$$-\frac{\partial^2 v_\theta}{\partial z^2} = \frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) + C_1$$

$$-\frac{\partial^2 v_\theta}{\partial z^2} = \frac{1}{r} v_\theta + \frac{C_1}{r} + C_2$$

$$\frac{C_1}{r} + C_2 = v_\theta \quad \text{D.C.}$$

$$C_1 = 0$$

$$C_2 = R\Omega$$

$$f(z) + \frac{\partial^2 v_\theta}{\partial z^2} = -\left(v_\theta + \frac{C_1}{r} + C_2 \right)$$

$$\frac{\partial^2 v_\theta}{\partial z^2} = v_\theta + R\Omega$$

separation of variables
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evaluate RHS

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$$\frac{d^2 V_\theta}{dz^2} = V_\theta + R\Omega$$

↓

$$\frac{dV_\theta}{dz} = \frac{V_\theta}{z} + \frac{R\Omega}{z} + C_1$$

$$V_\theta = -\frac{V_\theta}{z^2} - \frac{R\Omega}{z^2} + \frac{C_1}{z} + C_2$$

$$f(z) \text{ BL } \begin{cases} f(z=B) = 12.52 \\ f(z=0) = 0 \end{cases}$$

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